

Lesson 2 Logical Connectives

1. Let x be an integer. Consider the following statements:

p : “ $x \leq 4$ ”, q : “ $x = 4$ ”, r : “ x is even”, s : “ x is odd”

- Write a negation of each statement.
- Write $p \wedge \sim q$.
- Write $\sim p \vee q$.
- Write $r \rightarrow \sim s$.
- Write “ x is an even integer greater than 4” in terms of p, q, r , or s , with connectives.
- Write “ x is an odd integer less than 4” in terms of p, q, r , or s , with connectives.
- Determine the truth value of the following statement:
 - $q \wedge s$
 - $(q \wedge s) \rightarrow p$

2. We may define an exclusive OR, xor, denoted $\underline{\vee}$, as follows:

$$p \underline{\vee} q = p \leftrightarrow \sim q$$

Write the truth table for $p \underline{\vee} q$.

3. Write the converse and inverse of the following conditionals:

- If an integer is not even, then it is odd.
- If a number x is negative, then $|x| = -x$.
- If p is a prime number, then $\sqrt{p}$ is not rational.

4. True or false:

- If $a > b$ is true, then $a < b$ is false.
- If $a > b$ is false, then $a < b$ is true.
- If a statement p is false, then the conditional $p \rightarrow q$ is always true.
- If a statement p is false, then the biconditional $p \leftrightarrow q$ is always true.
- If a conditional is true, then its converse is true.
- If a conditional is true, then its converse is false.